EAST WEST UNIVERSITY

Course Code: CSE475

Course Title: Machine Learning

Section: 1

Project Title

Multiclass malware-family classification

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CSE475 Project — Task 1

Malimg Malware Image Dataset — Exploratory Data Analysis

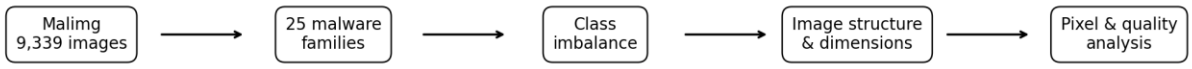


Figure 1. EDA workflow used for the Malimg dataset.

1. Dataset Overview

The Malimg dataset used in this project contains 9,339 malware images belonging to 25 malware families. The dataset is organized by malware family, where each family corresponds to a class. All 9,339 files were successfully read during the EDA, and every image was identified as a grayscale image (mode L).

The dataset is highly imbalanced. Allaple.A is the largest class with 2,949 images, whereas Skintrim.N is the smallest with 80 images. The resulting majority-to-minority ratio is approximately 36.86:1. This imbalance is an important characteristic of the dataset and should be considered when evaluating the later CNN + Attention model.

2. Malware Family Distribution

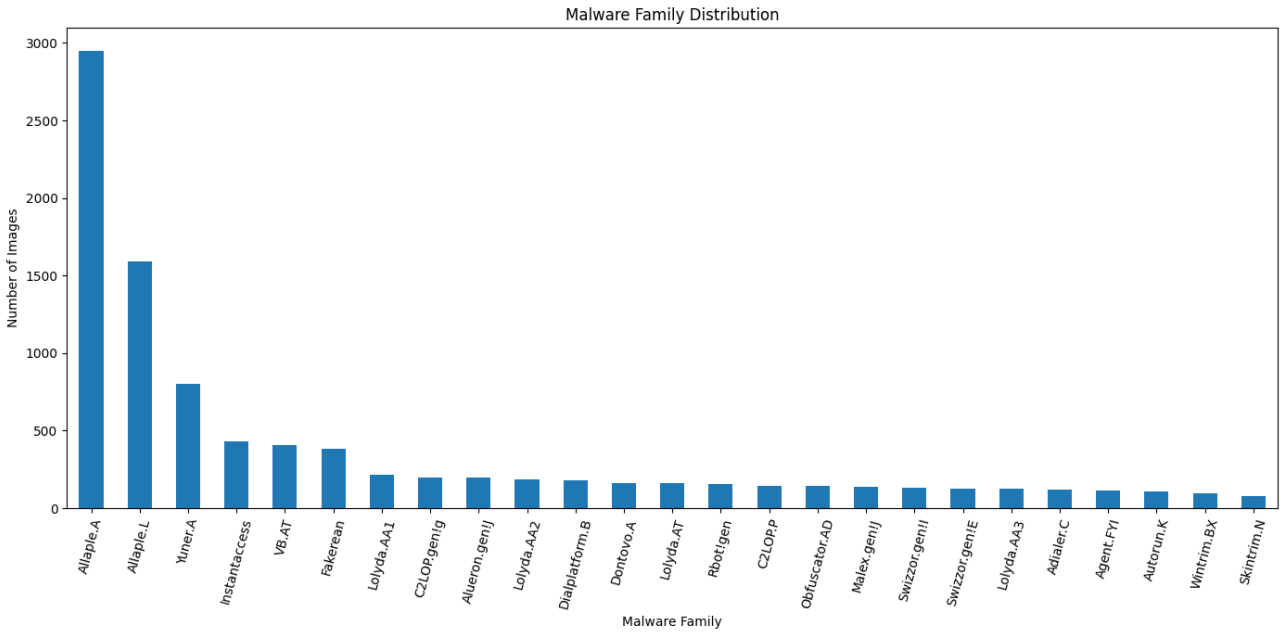


Figure 2. Distribution of images across the 25 malware families.

The distribution is strongly skewed toward a small number of families. Allaple.A and Allaple.L together account for a large portion of the dataset, while several families contain fewer than 150 images. Therefore, a model could obtain a

high overall accuracy while still performing poorly on minority classes. Per-class results and macro-averaged metrics will therefore be important in later evaluation.

3. Sample Image Inspection

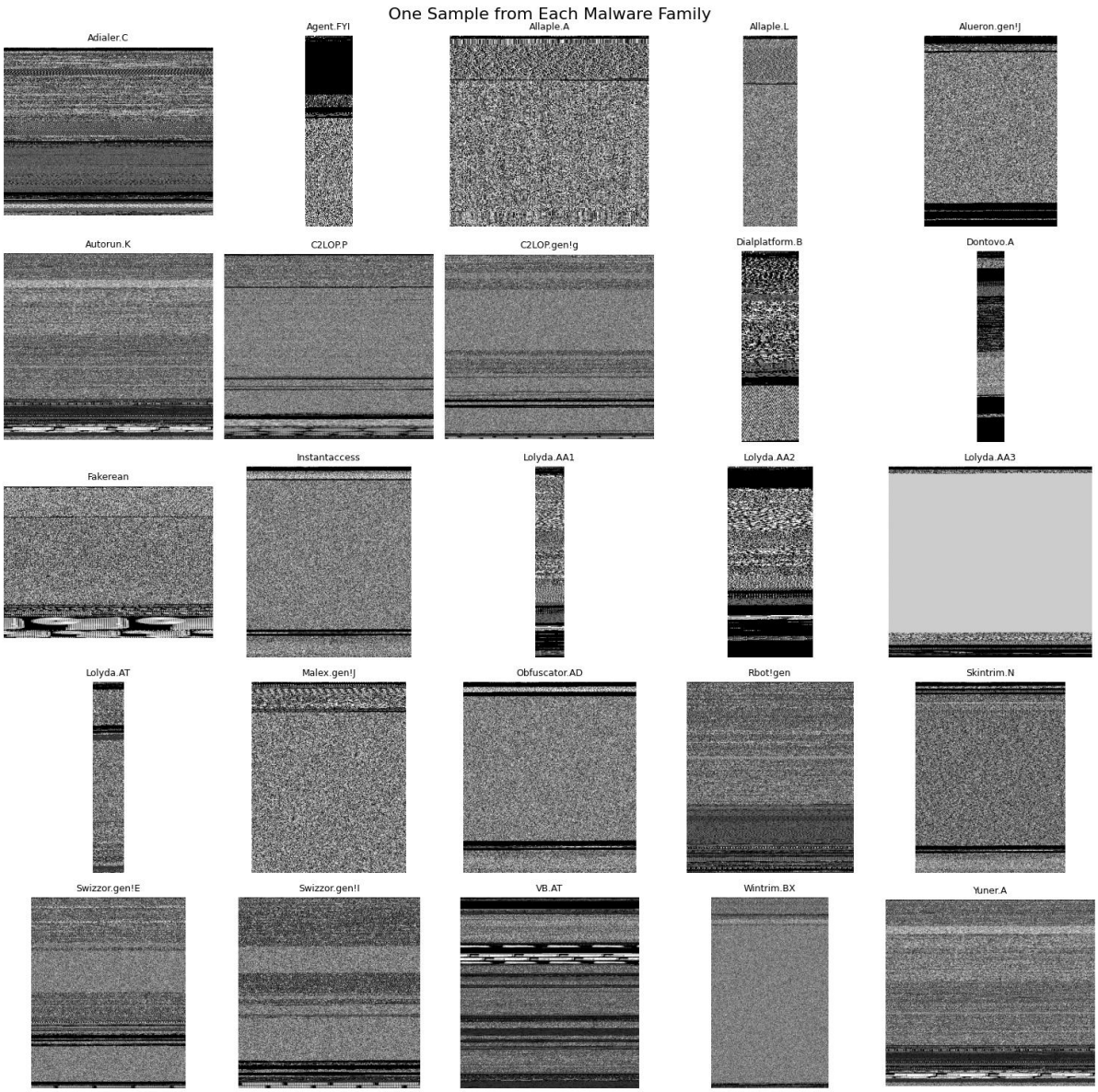


Figure 3. One sample image from each of the 25 malware families.

The sample grid provides a visual comparison of the malware families. The images are grayscale byte-image representations rather than natural photographs. Different families show different visual textures and spatial patterns, while some families can appear visually similar. These patterns motivate the use of convolutional feature extraction in the subsequent classification model.

4. Image Dimensions and Resolution

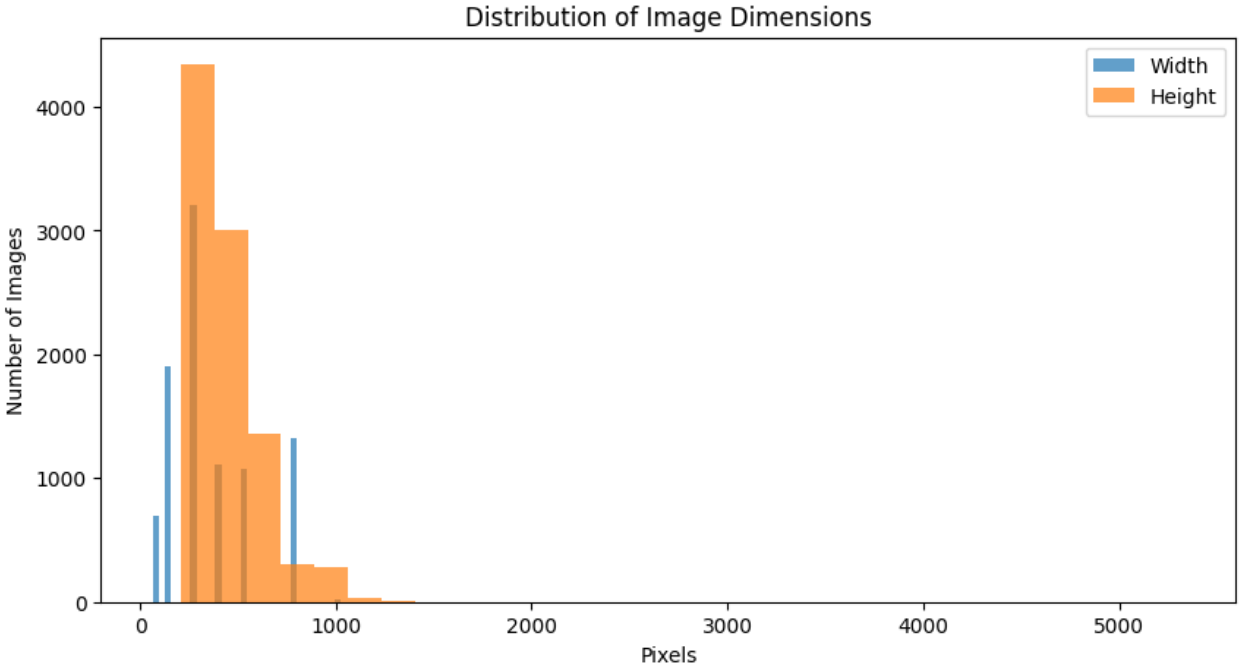


Figure 4. Distribution of image widths and heights in the dataset.

The original images do not share one fixed resolution. The width ranges from 64 to 1,024 pixels, with a mean of approximately 334.17 pixels. Heights range from 208 to 5,334 pixels, with a mean of approximately 436.81 pixels. The median width is 256 pixels and the median height is 424 pixels. This variation means that a fixed input size will be required during CNN preprocessing.

5. Pixel / Grayscale Analysis

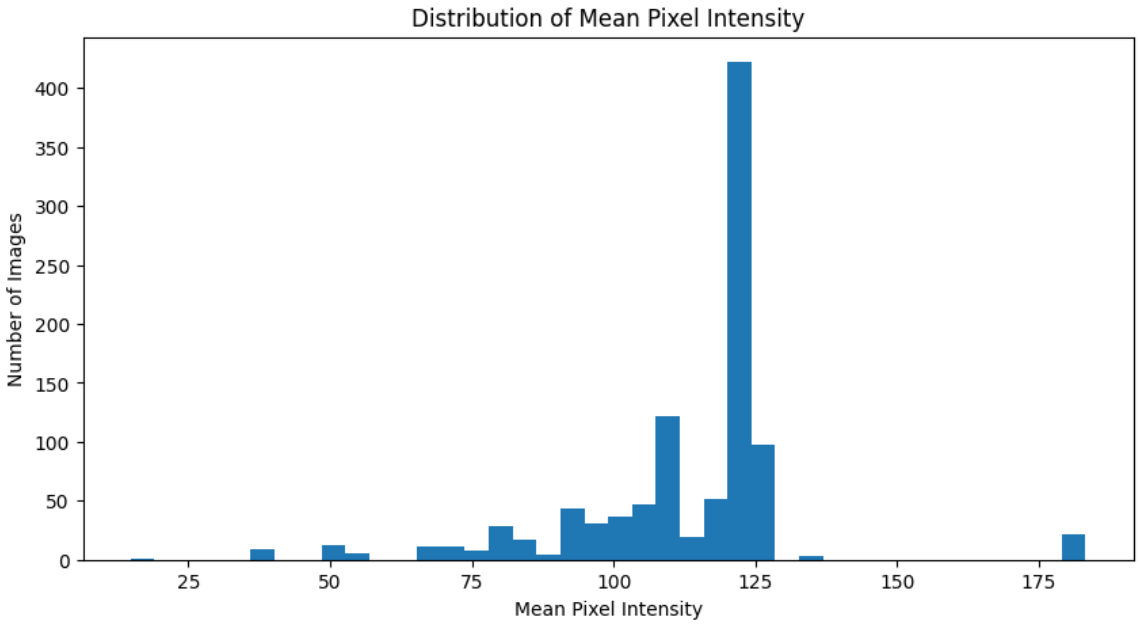


Figure 5. Distribution of mean pixel intensity for the sampled images.

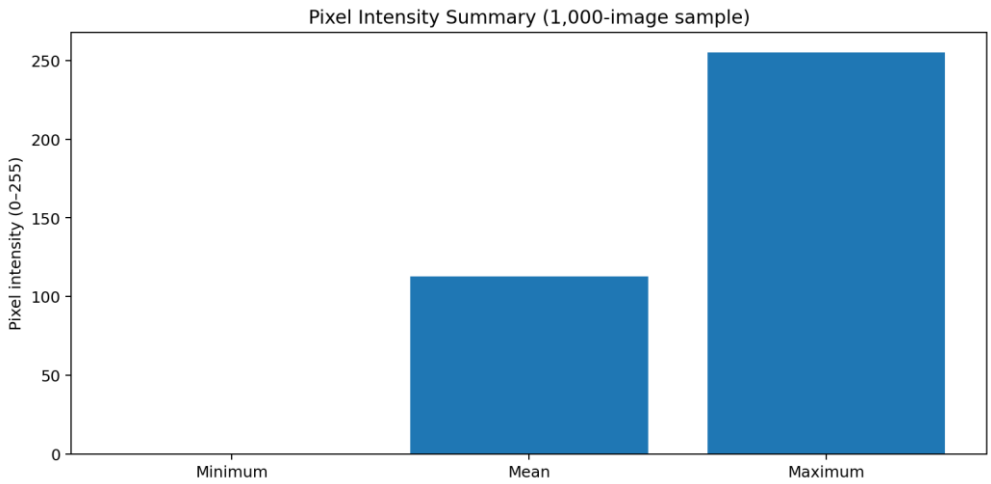


Figure 6. Summary of observed pixel intensity values.

A sample of 1,000 images was used for the pixel-statistics analysis. The mean pixel intensity was approximately 112.90, with an average within-image pixel standard deviation of approximately 79.40. The observed minimum and maximum intensities were 0 and 255, respectively. This confirms that the input is represented through the full 8-bit grayscale intensity range.

6. Image Quality Check

All 9,339 files passed the image verification step. There were 0 unreadable or corrupted images. Every image was successfully opened and verified, so no corrupted-file removal was required during this EDA. The dimension analysis also showed that the original images have substantial variation in size, which is relevant for later resizing and batching.

7. Overall EDA Findings

The Malimg dataset contains 9,339 grayscale malware byte-images distributed across 25 malware families. The class distribution is highly imbalanced: Allaple.A contains 2,949 samples whereas Skintrim.N contains only 80, resulting in a majority-to-minority ratio of approximately 36.86:1. Therefore, overall accuracy alone may hide poor performance on minority malware families. Macro-F1 and per-class recall should be emphasized during model evaluation.

9,339 malware images

25 malware families

Grayscale byte-image representation

Multiclass malware-family classification

Strong class imbalance

Largest class: Allaple.A — 2,949

Smallest class: Skintrim.N — 80

Majority/minority ratio: 36.86:1